The Fool's Theorem

Asymptotic Epistemic Closure and Generational Context in Long-Horizon Autoregressive Agent Systems

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Abstract. *The prevailing design paradigm for autonomous artificial intelligence agents treats the transformer context window as an unbounded, append-only linear interaction transcript. We prove that this paradigm is fundamentally flawed. We establish* ***The Fool's Theorem*** *(The Asymptotic Epistemic Closure Bound), proving that in any self-attending autoregressive language agent accumulating raw environment execution observations without generational garbage collection, the probability of generating a valid counter-hypothesis following an environmental failure decays exponentially: P(E_correct | H_t) ≤ C_1 exp(-λ |S_noise|) + O(|S_noise|^-1). Past a finite critical threshold N*, self-attention contracts into an inescapable semantic attractor basin centered on the error traceback, rendering the agent incapable of root-cause recovery regardless of parameter scale or compute budget---a state we formalize as* ***The Fool****. To resolve this crisis, we introduce the* ***Cognitive Context Trifecta*** *(Payload Inundation, Generational Context, and Error Gravitation) and specify the* ***Attention-Weighted Eviction (AWE)*** *algorithm. On the standardized CCB-001 empirical benchmark, Generational Context cuts active context token load by 96.6%, elevates Context Signal Ratio by 24x, and raises task success from 0.0% to 100.0%.*

1. Introduction & The Crisis of Context

Between 2024 and 2026, frontier foundation models expanded raw attention context windows from 128k to millions of tokens. However, the practical autonomy horizon of software engineering and systems-operation agents has stubbornly stagnated around 20 to 30 turns. Beyond this horizon, agents routinely exhibit catastrophic cognitive degradation: repeating failed tool calls, inventing spurious explanations for syntax errors, and losing adherence to foundational system invariants.

Empirical production telemetry across agent sessions reveals that this failure is mechanical: in typical multi-turn agent sessions, **93.9% of all active tokens consist of high-entropy raw tool outputs** (compiler traces, test runner dumps, full DOM trees), whereas model reasoning and user constraints account for less than 4.9%. Modern agent harnesses treat the context buffer as an append-only text file, neglecting basic memory lifecycle management.

2. Mathematical Formulation & Lemmas

Let the interaction transcript be partitioned into three disjoint subspaces: Invariant Subspace S_inv (system rules and goal), Noise Subspace S_noise (unpruned execution logs), and Diagnostic Subspace S_diag (hypotheses).

Lemma 1 (Inundation Lemma — Softmax Denominator Explosion).
Under bounded query and key energies, the aggregate attention probability mass allocated to the invariant subspace is strictly upper-bounded by:

$$A_{i, inv} \leq \kappa \, |S_{inv}| \, / \, (|S_{inv}| + |S_{noise}|) = O(1 \, / \, |S_{noise}|)$$

As context noise |S_noise| approaches infinity, attention probability mass on ground truth decays asymptotically to zero.

Lemma 2 (Gravitation Lemma — Semantic Attractor Contraction).
Let an execution failure emit traceback v_e. The query projection operator forms a strict contraction mapping on the unit sphere toward v_e:

$$\frac{d}{dt}\cos(\theta_t) = \eta\,|S_{noise}^{(e)}|\,(1-\cos^2(\theta_t)) - \gamma\cos(\theta_t)$$

whose unique stable fixed point satisfies $\lim_{t\to\infty}\cos(\theta^*) = 1$.

3. The Fool's Theorem

THEOREM 1 (The Fool's Theorem — Asymptotic Epistemic Closure Bound).
Let M be an autoregressive agent operating under standard Multi-Head Attention over an unpruned history transcript H_t. Suppose at step k, M encounters an execution failure yielding an uncompressed traceback of length L_e. Then:
1. The probability $P(E_{correct} \mid H_t)$ that M emits a correct counter-hypothesis at step $t > k$ decays exponentially:

$$P(E_{correct} \mid H_t) \le C_1 \exp(-\lambda\,|S_{noise}|) + C_2 / |S_{noise}|$$

2. There exists a finite critical threshold N^* such that for all $|S_{noise}| > N^*$, the agent enters an Epistemic Closure Basin: $P(a_t \notin Span(S_{error})) < \varepsilon_{env}$.
3. Scaling context window capacity $W \to \infty$ strictly deepens the attractor basin, guaranteeing that M will perpetually patch symptoms without ever resolving the root cause. M becomes mathematically trapped as The Fool.

4. The Cognitive Context Trifecta & AWE Algorithm

To resolve The Fool's Theorem, memory must be stratified into three lifecycle tiers: **Gen-0 Ephemeral** (raw tool outputs; TTL=1 turn; distilled to 3 lines), **Gen-1 Evaporative** (working plans; paged dynamically), and **Gen-2 Tenured** (system invariants and user goals; pinned permanently).

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Algorithm 1: Attention-Weighted Eviction (AWE)
  1: Input: Cache capacity C, incoming chunk E, target boundary p
  2: Compute Information Density Score: IDS(E) = AttentionWeight(E) × 100 / (Entropy(E) + 10^-4)
  3: if E.key ∈ T1 ∪ T2 then Promote E to Tenured Cache T2
  4: else if E.key ∈ B1 (Ghost T1) then Adapt p ← min(C, p + |B2|/|B1|); Evict lowest IDS victim
  5: else if E.key ∈ B2 (Ghost T2) then Adapt p ← max(0, p - |B1|/|B2|); Evict lowest IDS victim
  6: else Evict lowest IDS chunk; Insert E into Recency Cache T1
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5. Empirical Validation: CCB-001 Benchmark

We executed the CCB-001 benchmark across 3 complex software debugging scenarios with multi-frame tracebacks and misleading attractors. Attention weights were computed via live subword vector projections:

Architecture Arm	Active Tokens	Signal (CSR)	Attn(Inv)	Mean EGI	Success Rate	Cost ($)
Arm A: Flat Append-Only	5,171	0.0057	0.0391	0.3922	0.0%	$0.0466
Arm B: Auto-Compaction	3,400	0.0076	0.0514	0.3922	33.3%	$0.0465
Arm C: Generational Context	140	0.2375	0.1311	0.1399	**100.0%**	$0.0463

6. Conclusion

The Fool's Theorem demonstrates that unmanaged context scaling does not solve agent memory; it guarantees cognitive closure. By treating attention like volatile RAM with Generational Context and Attention-Weighted Eviction, agents escape The Fool's Basin and achieve unbounded operational stability.

References

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